Does Skill Abundance Still Matter?

The Evolution of Comparative Advantage in the 21st Century

Shin Kikuchi, UCSD

11/13/2026, NCSU

Skill Abundance and Comparative Advantage

- Skill Abundance: Central to comparative advantage (**Heckscher–Ohlin**)
  - Skill-abundant countries specialize in skill-intensive sectors

“Illustrating” the Idea of Heckscher–Ohlin

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United States



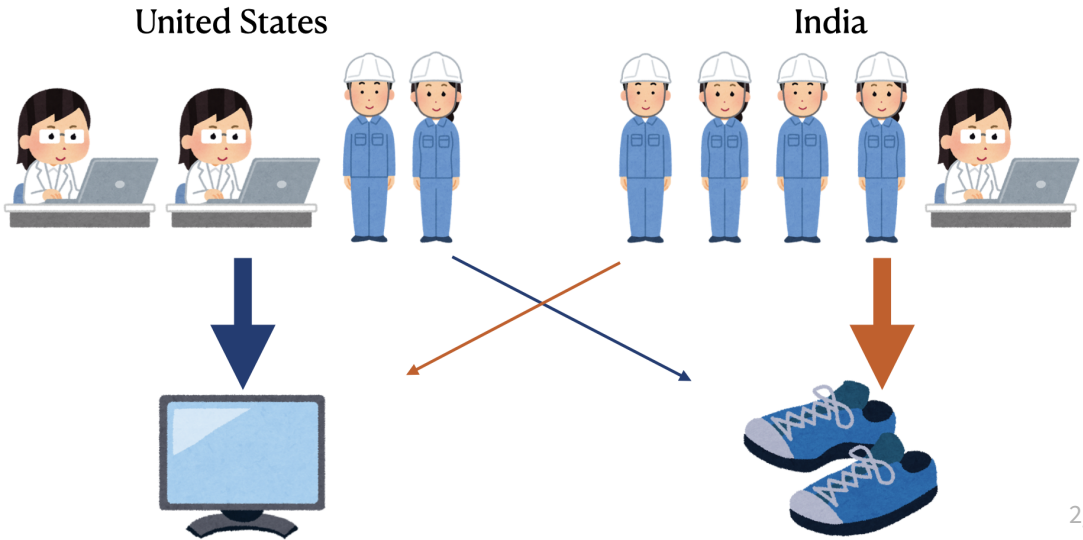
India







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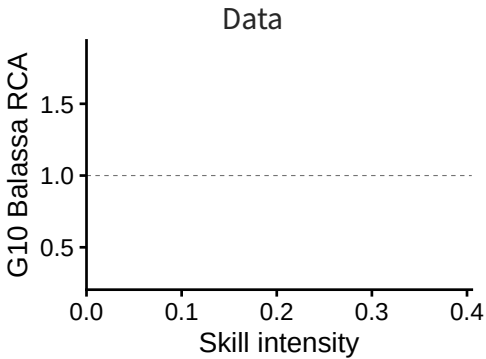
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At a Glance: Skill Abundance and Comparative Advantage

G10's Revealed Comparative Advantage (RCA) in skill-intensive sectors:

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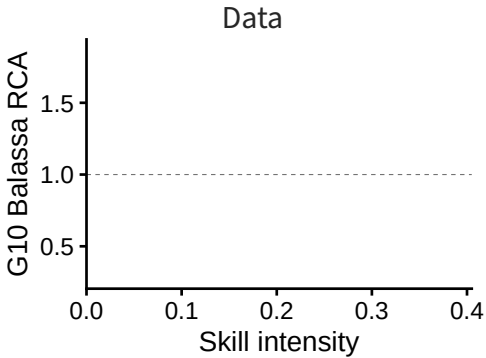
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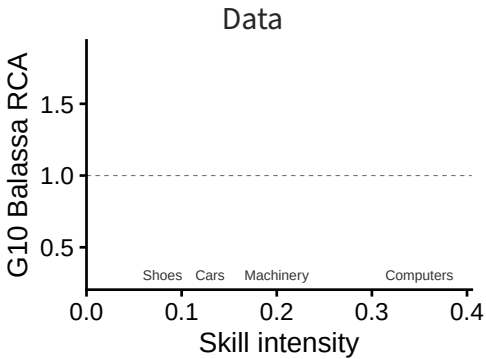


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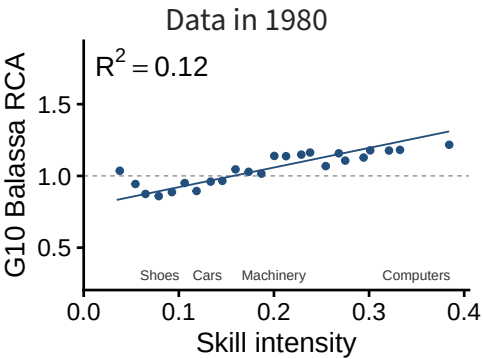
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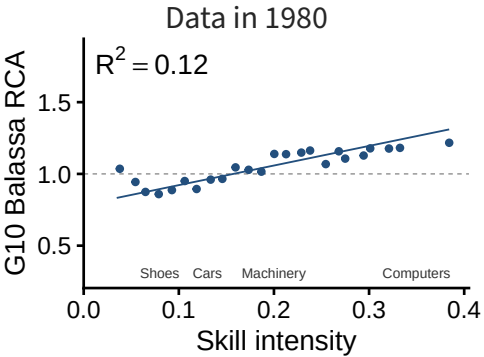
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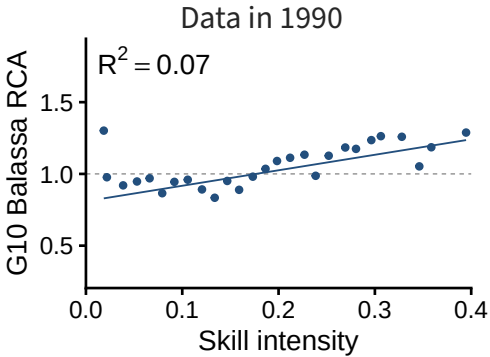
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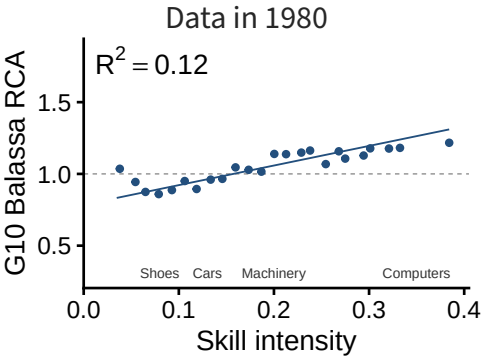


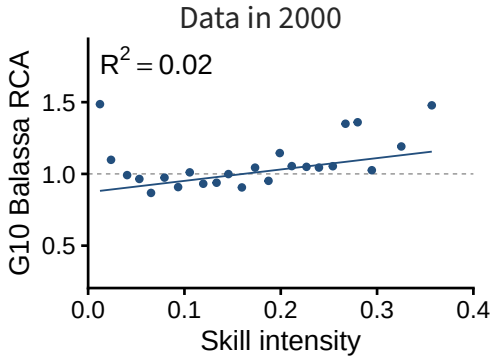


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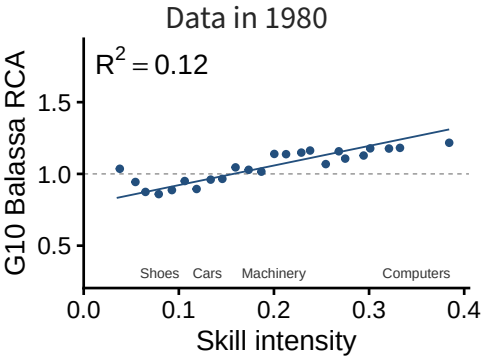


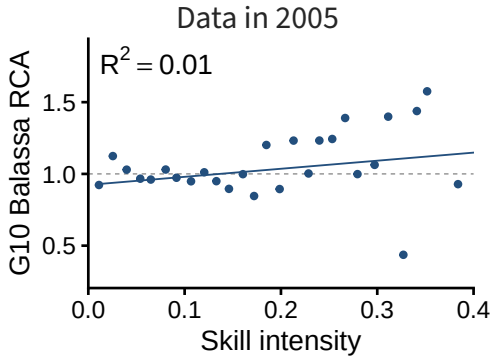


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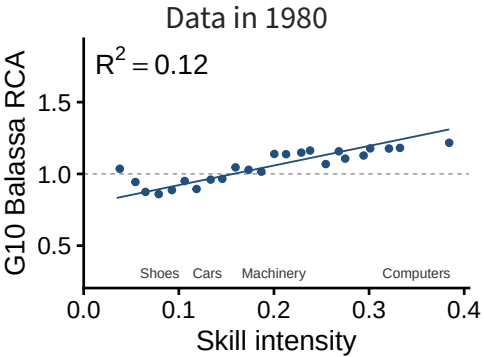


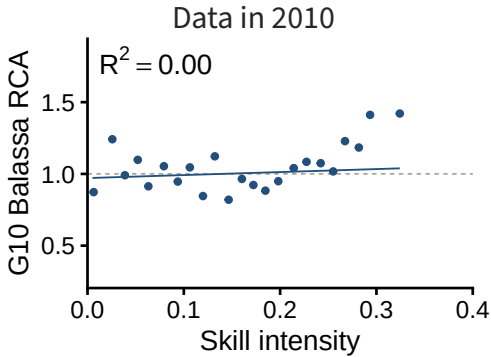


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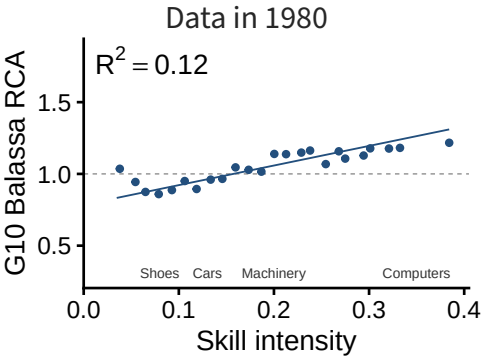


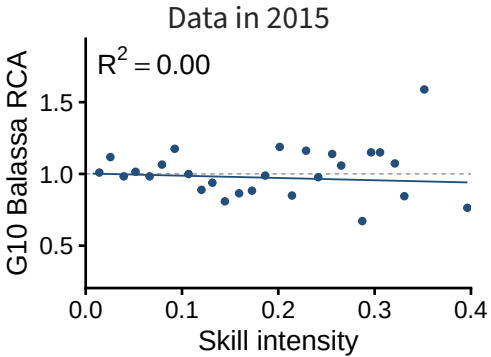


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   - Skill premia and welfare increase
   - **Lower trade costs amplify automation's effects**

Today's Plan

1. Skill Abundance as a Source of Comparative Advantage
2. Decomposing Changes in Comparative Advantage
3. Effects on Skill Premium and Welfare
4. Conclusion

SKILL ABUNDANCE AS A SOURCE OF COMPARATIVE ADVANTAGE

Identifying Comparative Advantage: Refresher

▸ Relative Wage

- Multi-sector Eaton–Kortum model (Chor (2010), Costinot et al. (2012))

Exporter i, importer j, sector s: $\ln \text{Exports}_{i,j,s} = -\underbrace{\theta}_{\text{Trade Elas.}} \underbrace{\ln c_{i,s}}_{\text{Unit Cost}} + \underbrace{\tilde{\eta}_{i,j} + \tilde{\eta}_{j,s}}_{\text{FEs}}$

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- Unit cost (α_s^H: skilled labor share of total payroll)

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- $\beta > 0$: **Skill-abundant countries specialize in skill-intensive sectors**

Skill Abundance as a Source of Comparative Advantage

Do skill-abundant countries export more skill-intensive goods?

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Skill Abundance as a Source of Comparative Advantage

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$$\mathbb{E}[X_{i,j,s,\mathbf{t}}] = \exp\left(\beta_{\mathbf{t}}\left[\alpha^{H}_{s,\mathbf{t}} \ln(H_{i,\mathbf{t}}/L_{i,\mathbf{t}})\right] + \eta_{i,j,\mathbf{t}} + \eta_{j,s,\mathbf{t}}\right)$$

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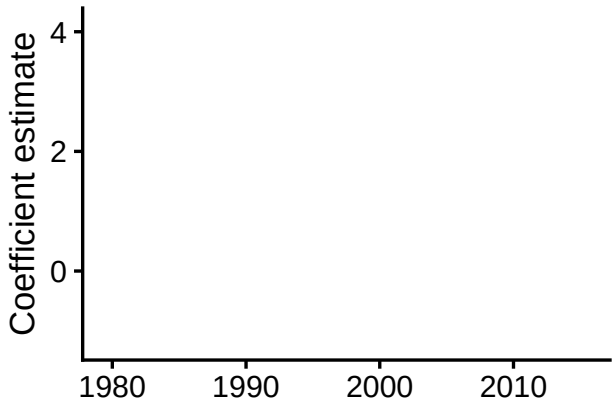
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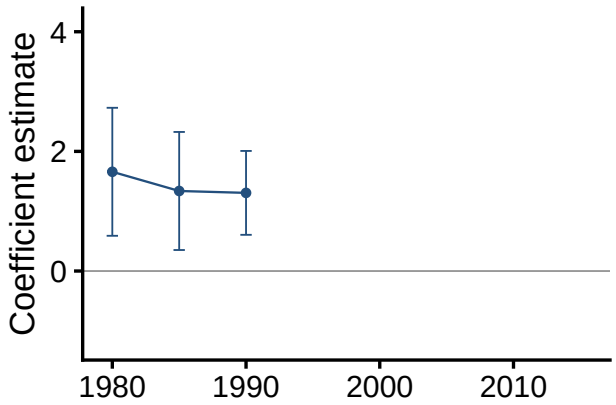
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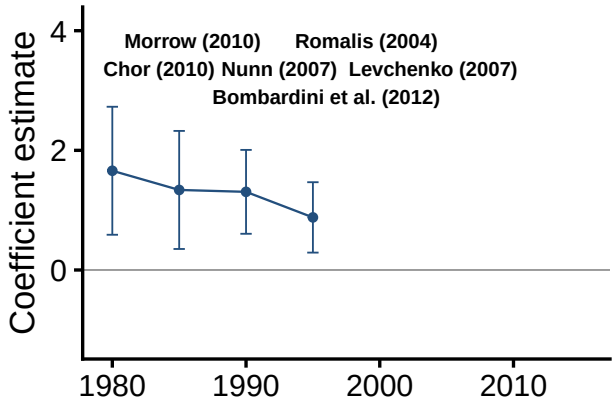
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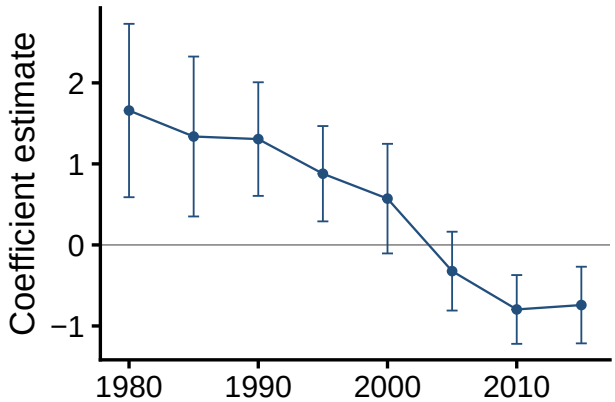
Early Results and Previous Research

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Declining Importance of Skill Abundance

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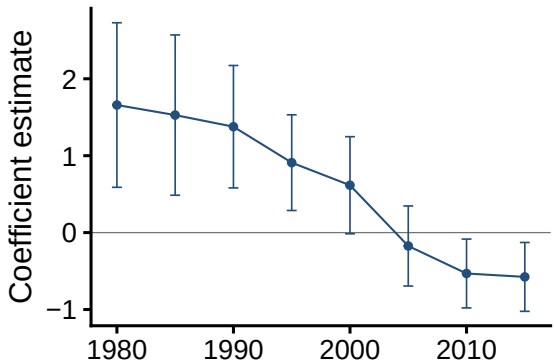
▸ Robustness Checks

Fixing Skill Abundance and Intensity at 1980 Values

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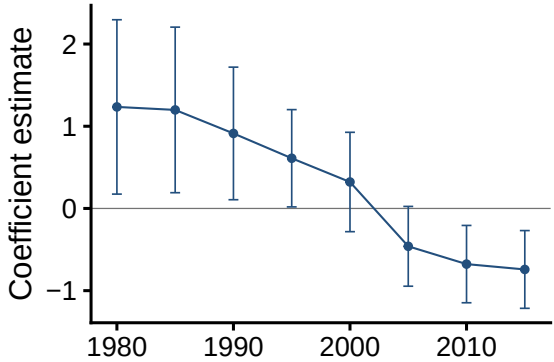


Fixing Skill Abundance and Intensity at 2015 Values

$$\mathbb{E}[X_{i,j,s,t}] = \exp\left(\beta_t \left[\alpha^H_{s,\mathbf{2015}} \ln(H_{i,\mathbf{2015}}/L_{i,\mathbf{2015}})\right] + \eta_{i,j,t} + \eta_{j,s,t}\right)$$

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More Robustness Checks

- Other sources of comparative advantage?
  - Capital ‣ go Institutions ‣ go
- Different specifications?
  - Log OLS ‣ go Exporter-sector fixed effects ‣ go
  - Total exports ‣ go
- Different measures of skill abundance and intensity?
  - Chor (2010) ‣ go Fixed inputs: 1980 ‣ go 2015 ‣ go
  - College payroll share, IPUMS ‣ go

Heterogeneity by Automation

Does the decline differ by automation across country-sectors?

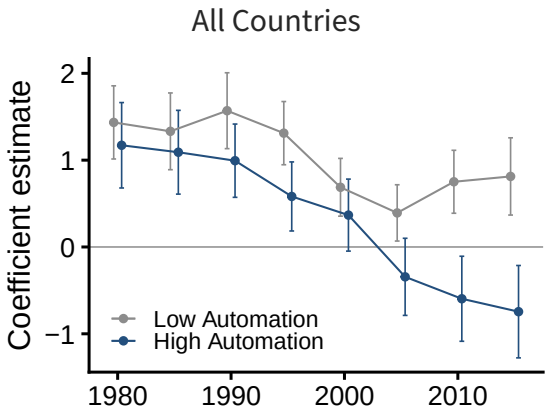
$$\mathbb{E}[X_{i,j,s,t}] = \exp\Big(\big(\beta_t^0 + \beta_t^A \, \mathbf{HA}_{i,s}\big) \big[\alpha_{s,t}^H \ln(H_{i,t}/L_{i,t})\big] + \gamma_t \, \mathbf{HA}_{i,s} \ln(H_{i,t}/L_{i,t}) + \eta_{i,j,t} + \eta_{j,s,t} \Big)$$

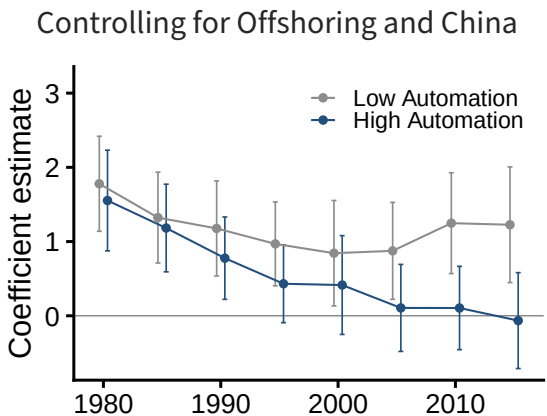
where

$$\mathbf{HA}_{i,s} \equiv \mathbf{1}\big\{ \text{Robots per worker}_{i,s} \text{ in the top 33\% of country} \times \text{sector-group pairs} \big\}$$

- Robots: IFR operational stock in 2015; workers: persons engaged in 1995, PWT
- 10 sector groups (2-digit SIC); 143 exporters, zero robots if not in IFR
- β_t^0: low automation; $\beta_t^0 + \beta_t^A$: high automation

More Decline in β in High-Automation Country-Sectors





Empirics: Taking Stock

- Skill abundance used to be a source of comparative advantage
- Not anymore after 2000
  - Interaction of skill abundance and intensity does not predict exports
- More decline in β in high-automation country-sectors
  - Skill abundance still matters in low-automation country-sectors

DECOMPOSING CHANGES IN COMPARATIVE ADVANTAGE

Trade Model with Automation

Setting

- Multi-sector Eaton–Kortum model
- Input-output linkages
- High-skilled labor H_{is}
- Tasks: low-skilled L_{is} or capital K_{is}

Gravity, $u \in \{G, M\}$: general, intermediate

$$\pi_{ijs}^{u} = \frac{\lambda_{is}(c_{is}\tau_{ijs}^{u})^{-\theta}}{\sum_{\ell} \lambda_{\ell s}(c_{\ell s}\tau_{\ell js}^{u})^{-\theta}}$$

‣ Full Model

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Unit cost

$$c_{is} = \Lambda_{is}(w_i^H)^{\alpha_{is}^H}(w_{is}^T)^{\alpha_{is}^T}(w_{is}^M)^{\alpha_{is}^M}$$

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Unit cost

$$c_{is} = \Lambda_{is}(w^H_i)^{\alpha^H_{is}}(w^T_{is})^{\alpha^T_{is}}(w^M_{is})^{\alpha^M_{is}}$$

Tasks: each goes to the cheaper factor

$$w^T_{is} = \exp\Big(\int_{\mathcal{T}^L_{is}} \ln \frac{w^L_i}{\psi^L_{is}(x)}\, dx + \int_{\mathcal{T}^K_{is}} \ln \frac{w^K_{is}}{\psi^K_{is}(x)}\, dx \Big)$$

- Lower w^K_{is} expands $\mathcal{T}^K_{is}$: automation

▸ Full Model

Quantitative Data and Calibration

- **1980 benchmark; shocks through 2014**
  - 25 economies: 24 named economies and the rest of the world
  - 401 sectors: 390 manufacturing and 11 nonmanufacturing sectors

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  - Capital-flow tables: Ding (2026)
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- **Elasticities**
  - Trade elasticity $\theta = 4$; task-assignment elasticity $\varepsilon = 1.4$
  - ε: elasticity of $\Gamma_{is}^{K}/\Gamma_{is}^{L}$ to w_i^L/w_{is}^K

‣ Exact Hat Algebra

Automation, Trade-Cost, and Offshoring Shocks

Automation: NBER-CES, 1980–2014

- $\widehat{z} = z'/z$: endpoint/base ratio
- $\widehat{\rho}_s$: equipment price relative to manufacturing

$\widehat{\lambda}_s = \widehat{\rho}_s^{-\theta}$ in equipment sectors

- Same $\widehat{\lambda}_s$ in all 25 economies
- No direct Ricardian effect on $\widehat{\beta}$

Trade costs: Head–Ries (2001), symmetric

$$\tau^G_{ijs} = \left(\frac{X^G_{iis} \cdot X^G_{jjs}}{X^G_{ijs} \cdot X^G_{jis}} \right)^{\frac{1}{2\theta}}$$

$$\tau^M_{ijs} = \left(\frac{X^M_{iis} \cdot X^M_{jjs}}{X^M_{ijs} \cdot X^M_{jis}} \right)^{\frac{1}{2\theta}}$$

Offshoring: additional wedge

$$\widehat{\chi}_{ijs} \equiv \frac{\widehat{\tau}^M_{ijs}}{\widehat{\tau}^G_{ijs}}$$

Summary Statistics for the Calibrated Shocks

Summary Statistics (percentiles) for Shocks

Shock		p_{10}	p_{50}	p_{90}
Automation	$\widehat{\rho}_s$	0.40	0.98	1.28
Trade costs	$\widehat{\tau}_{ijs}$	0.30	0.66	1.25
Offshoring	$\widehat{\chi}_{ijs}$	0.56	1.03	2.09
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- Automation: relative equipment-price changes in 84 sectors
- Trade-cost/offshoring percentiles exclude diagonal and unit cells

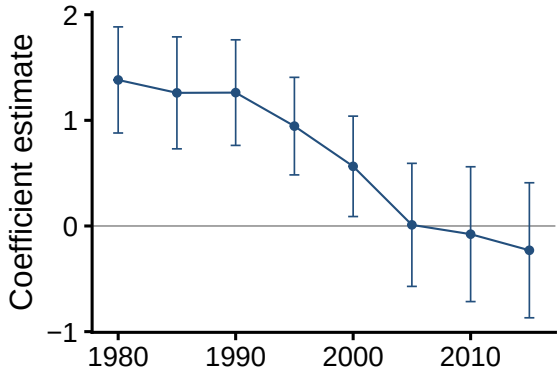
Empirics using the Model's Sample

Sample

- 24 economies; Hong Kong in China
- Cross-border manufacturing; no ROW

Specification

- Unweighted PPML
- Skill interaction fixed at 1980
- Model shocks: 1980–2014
- Data: 2015 wave (2014–16 avg.)



Decomposition via Counterfactual Combinations

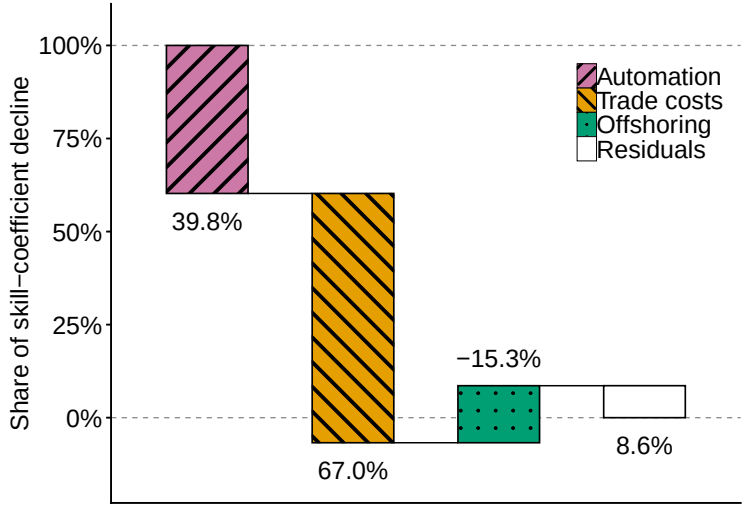
- **Three shock blocks**
  - Automation, trade costs, and offshoring
  - Solve all 2^3 combinations from the 1980 equilibrium
    - ⋆ Shapley decomposition (invariant across decomp. orders)

Decomposition via Counterfactual Combinations

- **Three shock blocks**
  - Automation, trade costs, and offshoring
  - Solve all 2^3 combinations from the 1980 equilibrium
    - ⋆ Shapley decomposition (invariant across decomp. orders)
- **Counterfactual regressions**
  - Endowments and deficit shares remain fixed
    - ⋆ RHS of the regression is fixed
  - Re-estimate PPML on each counterfactual trade matrix

$$\mathbb{E}[X_{i,j,s,t}^{\text{Model}}] = \exp\left(\beta_t \left[\alpha_{s,1980}^{H} \ln(H_{i,1980}/L_{i,1980})\right] + \eta_{i,j,t} + \eta_{j,s,t}\right)$$

Decomposing Changes in Comparative Advantage



EFFECTS ON SKILL PREMIUM AND WELFARE

Macroeconomic Counterfactuals

- **Counterfactual assumptions**
  - Measured 1980–2014 skill-supply changes in every experiment
  - Intermediate-use wedges and deficit shares fixed at 1980

Macroeconomic Counterfactuals

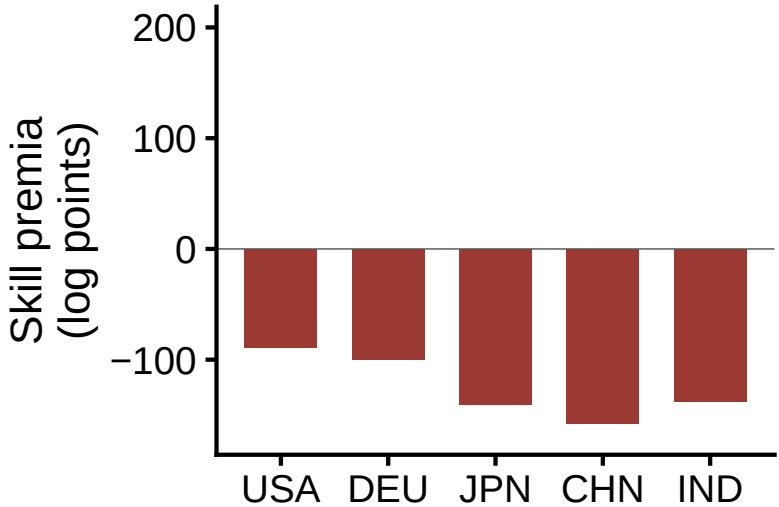
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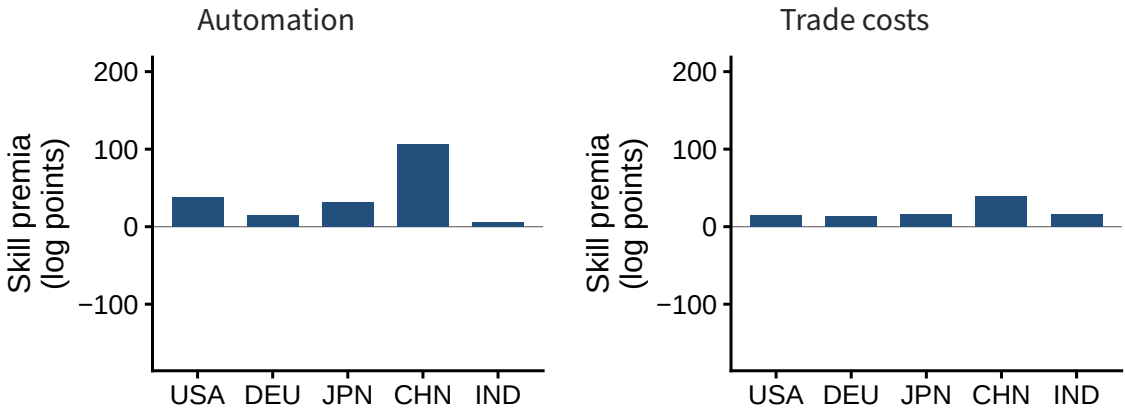
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- **Shock effects** (y_i^Y: log-point outcome relative to 1980)

$$y_i^Y(a, t) - y_i^Y(0, 0), \qquad Y \in \{\text{SP}, W\}$$

Endowment Changes Alone: Relative to 1980



The Race for Skills: Automation and Trade Costs



Both shocks raise the skill premium in every economy.

Skill-Premium Accounting

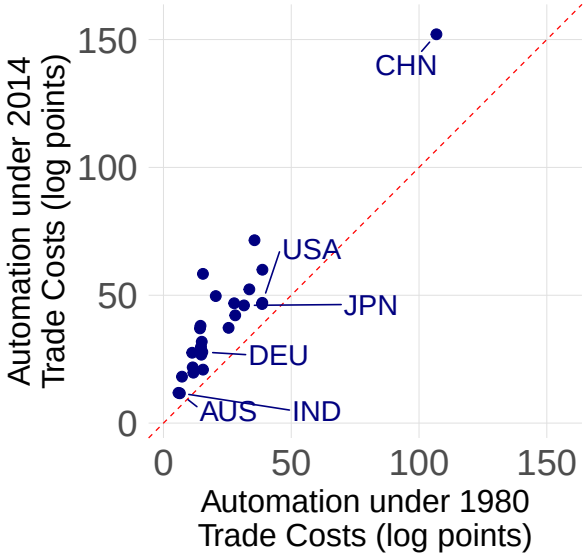
- **Mean effects across 25 economies; log points**
  - First three: relative to endowments alone; last two: relative to 1980

Shocks	Supply	Sec. Reallocation	Task	Total
Automation	0.00	−2.49	26.43	23.94
Trade costs	0.00	−6.00	30.20	24.20
Both Shocks	0.00	−9.10	74.61	65.51
Endowment Only	−153.12	−1.34	15.81	−138.65
Both Shocks + Endowment	−153.12	−10.44	90.41	−73.15

- Task reassignment raises skill premia; sectoral reallocation partly offsets it

▸ Details

Lower Trade Costs Amplify Automation: Skill Premia



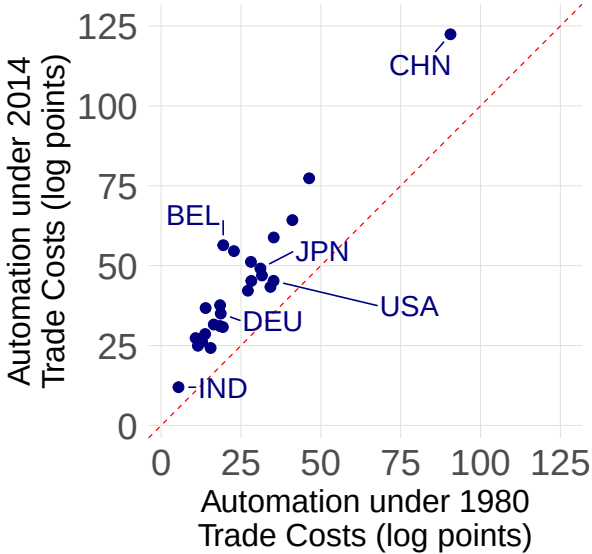
Welfare Effects

- **Across 25 economies.**
  - First three: relative to endowments alone; last two: relative to 1980

Shocks	Min	Median	Max
Automation	5.4 (IND)	19.5	90.6 (CHN)
Trade costs	16.9 (AUS)	39.0	100.6 (BEL)
Both Shocks	29.3 (IND)	71.9	164.7 (CHN)
Endowment Only	7.1 (JPN)	39.7	126.4 (MEX)
Both Shocks + Endowment	80.1 (FIN)	132.4	242.0 (CHN)

‣ Real Income

Lower Trade Costs Amplify Automation: Welfare



CONCLUSION

Conclusion

- **Skill abundance no longer predicts skill-intensive specialization**
  - More decline in β in high-automation country-sectors
  - Skill abundance still matters in low-automation country-sectors
  - Lower trade costs and automation explain most of the decline
  - Market access accounts for most of the trade-cost contribution

- **Lower trade costs amplify automation's effects**
  - Task reassignment raises skill premia
  - Welfare increases

APPENDIX

Key Contributions

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   - **Ricardian:** MacDougall (1951), Stern (1962), Balassa (1963), Golub & Hsieh (2000), Nunn (2007), Levchenko (2007), Manova (2008), Costinot (2009), **Costinot et al. (2012)**
   - **HO:** Leamer (1980, 1984), Bowen et al. (1987), Trefler (1993, 1995), Harrigan (1997), Davis & Weinstein (2001), Schott (2001), **Romalis (2004)**, Morrow (2010), **Chor (2010)**
   - **Dynamics:** Levchenko & Zhang (2016), Hanson, Lind, and Muendler (2015)

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$\rightarrow$ **Skill abundance mattered in the 1980s; not anymore after 2000.**

Key Contributions

1. New facts on the sources of comparative advantage:
2. **Consequences of technology and globalization for inequality:**

Key Contributions

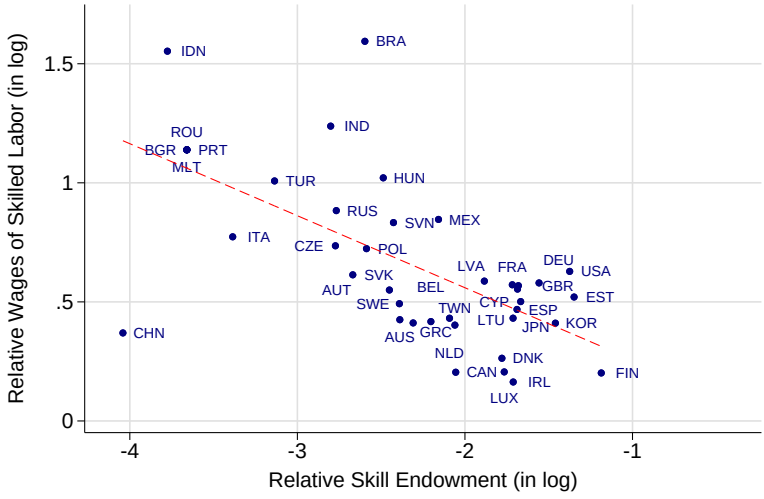
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   - **Offshoring: Feenstra & Hanson (1997, 1999, 2001), Grossman & Rossi-Hansberg (2008, 2012)**, Senses (2010), Hummels et al. (2014), Boehm et al. (2020)
   - **Interaction of technology and trade:** Xu (2001), Acemoglu (2003), Thoenig & Verdier (2003), Burstein et al. (2013), Parro (2013), Burstein & Vogel (2017), Morrow & Trefler (2022)
   - **Automation and trade:** Freund et al. (2022), Artuc et al. (2023), Fontagné et al. (2024)

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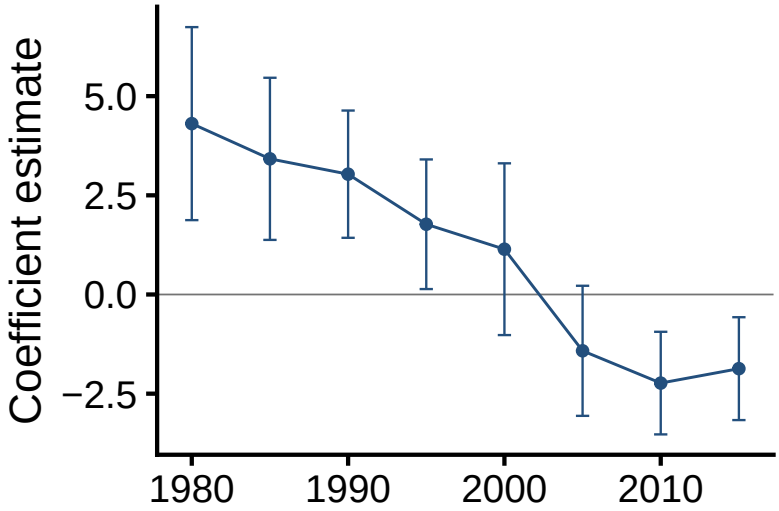
$\rightarrow$ **Automation** $\Rightarrow$ **Comparative advantage and inequality**

Relative Skilled Wages and Skill Endowment



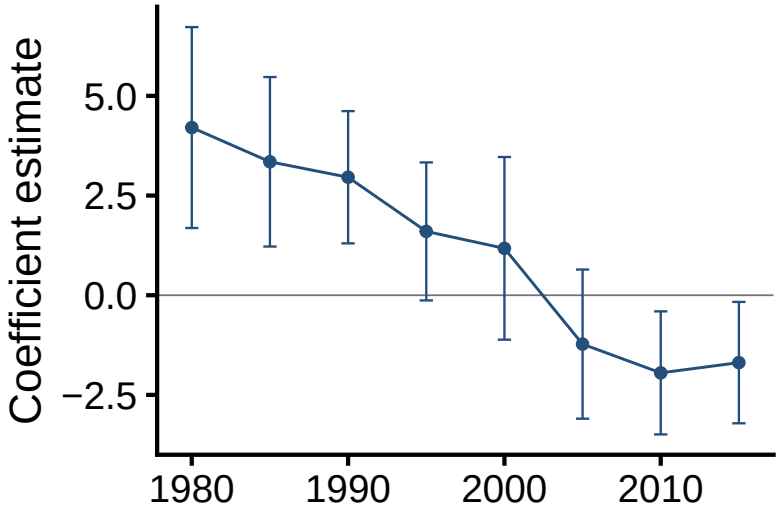
‣ Back

Controlling for Capital Intensity



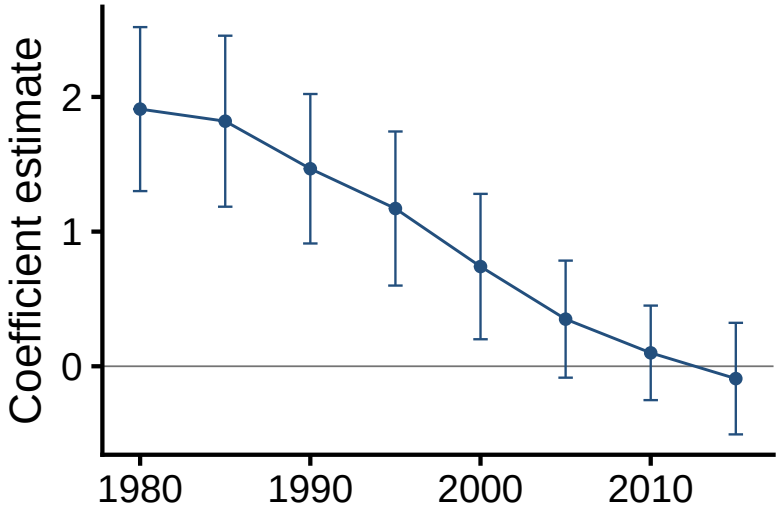
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Controlling for Capital Intensity and Institutions



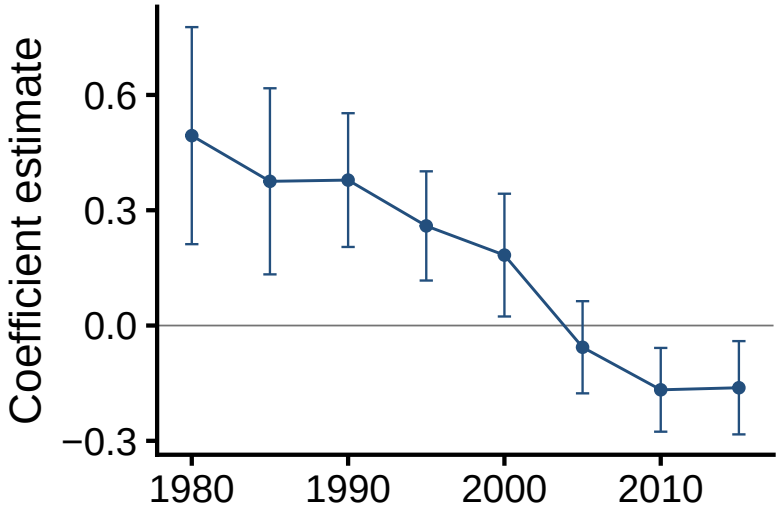
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Log OLS Specification



‣ Back

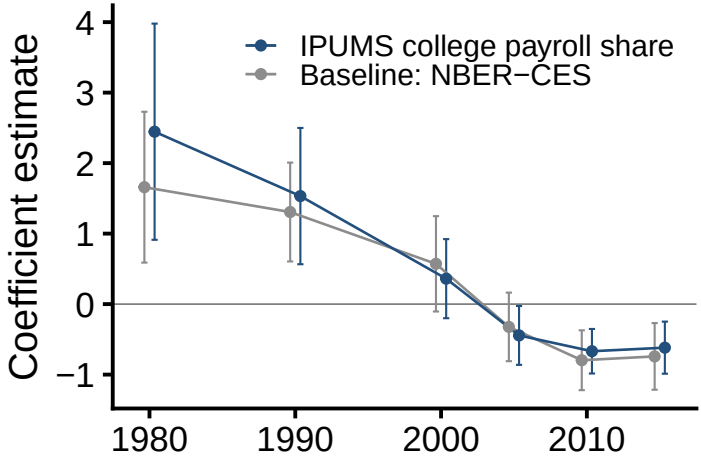
Chor (2010): Log Non-Production / Production Employment



‣ Back

Skill Intensity from IPUMS: College Payroll Share

$\alpha^{H}_{s,t}$: college (BA+) payroll / total payroll, IPUMS USA (Census 1980–2000; ACS 2004–16)



‣ Back

Heterogeneity by Automation: Adding Controls

$$\mathbb{E}[X_{i,j,s,t}] = \exp\Big(\big(\beta_t^0 + \beta_t^A \, \mathbf{HA}_{i,s} + \beta_t^O \, \text{Ofs}_{i,s} + \beta_t^C \, \text{RCA}_{s,t}^{CHN}\big) \big[\alpha_{s,t}^H \ln(H_{i,t}/L_{i,t})\big] + \big(\gamma_t \, \mathbf{HA}_{i,s} + \gamma_t^O \, \text{Ofs}_{i,s} + \gamma_t^C \, \text{RCA}_{s,t}^{CHN}\big) \ln(H_{i,t}/L_{i,t}) + \eta_{i,j,t} + \eta_{j,s,t} \Big)$$

$$\mathbf{HA}_{i,s} \equiv \mathbf{1}\big\{\text{Robots per worker}_{i,s} \text{ in the top 33\% of country} \times \text{sector-group pairs}\big\}$$

- Controls for offshoring and the rise of China:
  - $\text{Ofs}_{i,s}$: imported share of manufacturing intermediates in 2010, WIOD
  - $\text{RCA}_{s,t}^{CHN}$: China's revealed comparative advantage in sector s
- β_t^0: low automation; $\beta_t^0 + \beta_t^A$: high automation

‣ Back

Model: Overview

- Multi-sector Eaton–Kortum model with input-output linkages

▸ Full Model

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- Multi-sector Eaton–Kortum model with input-output linkages
  - **New: Adding a task framework for automation**
    - ⋆ Zeira, Acemoglu–Restrepo
- Demand: Cobb–Douglas across sectors, CES (EK) within sectors, gravity

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- Multi-sector Eaton–Kortum model with input-output linkages
  - **New: Adding a task framework for automation**
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- Demand: Cobb–Douglas across sectors, CES (EK) within sectors, gravity
- Supply: Multiple production factors
  - Primary: labor H_{is} (high-skilled), L_{is} (low-skilled)
  - Additional inputs produced using sectoral outputs (roundabout)
    - ⋆ Automation capital: K_{is}
    - ⋆ Intermediates: $M_{i,i,r,s}$ (domestic), $M_{-i,i,r,s}$ (foreign, offshored)

▸ Full Model

Demand: Standard Multi-Sector Eaton–Kortum Model

- Country i, j, sector s
- Preference across sectors: Cobb–Douglas with expenditure share ν_{js}

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$$\pi^{u}_{ijs} = \frac{\lambda_{is}(c_{is}\tau^{u}_{ijs})^{-\theta}}{\sum_{\ell \in \mathcal{I}} \lambda_{\ell s}(c_{\ell s}\tau^{u}_{\ell js})^{-\theta}}$$

  - Productivity scale λ_{is}: exogenous shifter
  - Unit cost c_{is}: endogenously determined from production
  - Iceberg trade costs: $\tau^{G}_{ijs} = \tau_{ijs}$, $\tau^{M}_{ijs} = \tau_{ijs}\chi_{ijs}$
  - Trade elasticity $\theta > 0$

Supply 1/2: Task Framework in Production

- Eaton–Kortum with a task framework

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$$q_{is}(\omega) = Z_{is}(\omega) \cdot (H_{is}(\omega))^{\alpha_{is}^{H}} \cdot (T_{is}(\omega))^{\alpha_{is}^{T}} \cdot (M_{is}(\omega))^{\alpha_{is}^{M}}.$$

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$$q_{is} = Z_{is} \cdot (H_{is})^{\alpha_{is}^{H}} \cdot (T_{is})^{\alpha_{is}^{T}} \cdot (M_{is})^{\alpha_{is}^{M}}.$$

  - Cost shares: α_{is}^{H} high-skilled labor; α_{is}^{T} tasks
  - α_{is}^{M}: intermediates; $\alpha_{is}^{H} + \alpha_{is}^{T} + \alpha_{is}^{M} = 1$

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- Eaton–Kortum with a task framework
- Gross Output

$$q_{is} = Z_{is} \cdot (H_{is})^{\alpha_{is}^{H}} \cdot (T_{is})^{\alpha_{is}^{T}} \cdot (M_{is})^{\alpha_{is}^{M}}. \quad T_{is} = \exp\left(\int_0^1 \ln \underbrace{T_{is}(x)}_{\text{Task}} \, dx \right).$$

- Task-specific productivities: $\psi_{is}^{L}(x)$ and $\psi_{is}^{K}(x)$

$$T_{is}(x) = \underbrace{\psi_{is}^{L}(x) \cdot L_{is}(x)}_{\text{Low-skilled labor}} + \underbrace{\psi_{is}^{K}(x) \cdot K_{is}(x)}_{\text{Capital}}$$

Supply 2/2: Task Allocation $\mathcal{T}_{is}^{f}$, Task Share Γ_{is}^{f}

- w_i^L: low-skilled wage; w_{is}^K: capital-service price
- Task-specific costs for $f \in \{L, K\}$:

$$\widetilde{w}_{is}^{L}(x) = \frac{w_i^L}{\psi_{is}^{L}(x)}, \qquad \widetilde{w}_{is}^{K}(x) = \frac{w_{is}^K}{\psi_{is}^{K}(x)}$$

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- Cost minimization determines task sets and their shares:

$$\mathcal{T}_{is}^{f} = \left\{ x : f = \underset{g \in \{L,K\}}{\operatorname{argmin}} \widetilde{w}_{is}^{g}(x) \right\}, \qquad \Gamma_{is}^{f} = \int_{\mathcal{T}_{is}^{f}} dx$$

Supply 2/2: Task Allocation $\mathcal{T}_{is}^{f}$, Task Share Γ_{is}^{f}

- Task allocation determines the task-composite price:

$$w_{is}^{T} = \exp\left(\int_{\mathcal{T}_{is}^{L}} \ln \widetilde{w}_{is}^{L}(x)\, dx + \int_{\mathcal{T}_{is}^{K}} \ln \widetilde{w}_{is}^{K}(x)\, dx \right)$$

- Task shares Γ_{is}^{f} set the domains in w_{is}^{T}
- w_{is}^{M}: intermediate-bundle price; Λ_{is}: cost-share constant
- Sectoral unit cost:

$$c_{is} = \Lambda_{is} (w_{i}^{H})^{\alpha_{is}^{H}} (w_{is}^{T})^{\alpha_{is}^{T}} (w_{is}^{M})^{\alpha_{is}^{M}}$$

Equilibrium Conditions

Given factor endowments $\{H_i, L_i\}$, an equilibrium is a set of wages $\{w_i^H, w_i^L\}$

- Household and firm sourcing yields π_{ijs}^u from unit costs $\{c_{\ell s}\}$
- Unit cost c_{is} depends on $\{w_i^H, w_i^L\}$
  - $\{w_{is}^K, w_{is}^M\}$ also depend on wages through input-output linkages
- Goods and labor markets clear

Model (1/7): Gross Production

- **Gross production**
  - i: country; s: sector
  - $H_{is}(\omega)$: high-skilled labor; $T_{is}(\omega)$: tasks; $M_{is}(\omega)$: intermediate inputs

$$q_{is}(\omega) = Z_{is}(\omega) \cdot [H_{is}(\omega)]^{\alpha_{is}^{H}} \cdot [T_{is}(\omega)]^{\alpha_{is}^{T}} \cdot [M_{is}(\omega)]^{\alpha_{is}^{M}}$$

▸ Back

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$$\alpha_{is}^H + \alpha_{is}^T + \alpha_{is}^M = 1$$

‣ Back

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$$\alpha_{is}^H + \alpha_{is}^T + \alpha_{is}^M = 1$$

- **Fréchet productivity** (λ_{is}: scale; θ: trade elasticity)

$$\Pr[Z_{is}(\omega) \leq z] = \exp(-\lambda_{is} z^{-\theta})$$

▸ Back

Model (2/7): Tasks

- **Task composite**
  - Tasks $x \in [0, 1]$ receive equal Cobb–Douglas weights

$$T_{is}(\omega) = \exp\left(\int_0^1 \ln T_{is}(\omega, x)\, dx\right)$$

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- **Production of each task**
  - $\psi_{is}^{L}(x)$ and $\psi_{is}^{K}(x)$: task-specific productivities

$$T_{is}(\omega, x) = \underbrace{\psi_{is}^{L}(x) \cdot L_{is}(\omega, x)}_{\text{Low-skilled labor}} + \underbrace{\psi_{is}^{K}(x) \cdot K_{is}(\omega, x)}_{\text{Capital}}$$

▸ Back

Model (3/7): Task Costs and Assignment

- **Task-specific unit costs**
  - $w_i^{H,L}$: high-/low-skilled wages; w_{is}^K: capital-service price

$$\widetilde{w}_{is}^L(x) = \frac{w_i^L}{\psi_{is}^L(x)} \qquad \widetilde{w}_{is}^K(x) = \frac{w_{is}^K}{\psi_{is}^K(x)}$$

‣ Back

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- **Automation**
  - Task x goes to capital when

$$\frac{\psi_{is}^K(x)}{\psi_{is}^L(x)} > \frac{w_{is}^K}{w_i^L}$$

  - Lower capital-service prices expand capital's task set

‣ Back

Model (4/7): Capital, Intermediate Inputs, and Unit Costs

- **Capital services/intermediate inputs**
  - $P_{ir}^{G,M}$: general/intermediate prices; κ, γ: capital/input weights

$$w_{is}^{K} = \prod_{r} (P_{ir}^{G})^{\kappa_{i,rs}}, \quad w_{is}^{M} = \prod_{r} (P_{ir}^{M})^{\gamma_{i,rs}}$$

- **Task-composite price** ($a_{is}(x)$: minimum cost of task x)

$$w_{is}^{T} = \exp\left(\int_{0}^{1} \ln a_{is}(x)\, dx \right)$$

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$$w_{is}^{T} = \exp\left(\int_{0}^{1} \ln a_{is}(x)\, dx \right)$$

- **Sectoral unit cost** (Λ_{is}: production-share constant)

$$c_{is} = \Lambda_{is} (w_{i}^{H})^{\alpha_{is}^{H}} (w_{is}^{T})^{\alpha_{is}^{T}} (w_{is}^{M})^{\alpha_{is}^{M}}$$

‣ Back

Model (5/7): General- and Intermediate-Use Sourcing

- **General use**
  - π^{G}_{ijs}: exporter *i*'s share; τ_{ijs}: iceberg trade cost

$$\pi^{G}_{ijs} = \frac{\lambda_{is}(c_{is}\tau_{ijs})^{-\theta}}{\sum_{\ell}\lambda_{\ell s}(c_{\ell s}\tau_{\ell js})^{-\theta}}$$

‣ Back

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- **Intermediate use**
  - χ_{ijs}: additional delivered-cost wedge

$$\pi^{M}_{ijs} = \frac{\lambda_{is}(c_{is}\tau_{ijs}\chi_{ijs})^{-\theta}}{\sum_{\ell}\lambda_{\ell s}(c_{\ell s}\tau_{\ell js}\chi_{\ell js})^{-\theta}}$$

▸ Back

Model (6/7): Trade Flows

- **Expenditure and trade**
  - E^u_{js}: expenditure in country j, sector s, use u
  - $u \in \{G, M\}$: general or intermediate use

$$X^u_{ijs} = \pi^u_{ijs} E^u_{js}$$

‣ Back

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- **Total trade flows**

$$X_{ijs} = X^G_{ijs} + X^M_{ijs}$$

  - General use: consumption and capital purchases
  - Offshoring: foreign share of intermediate expenditure

‣ Back

Model (7/7): Household Income and Expenditure

- **Households**
  - d_i: deficit share; V^W: world factor income
  - Cobb–Douglas preferences; ν_{is}: expenditure share

$$I_i = w_i^H H_i + w_i^L L_i + d_i V^W$$

‣ Back

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  - d_i: deficit share; V^W: world factor income
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$$I_i = w_i^H H_i + w_i^L L_i + d_i V^W$$

$$\underbrace{E_{is}^F}_{\text{Final}} = \nu_{is} I_i, \qquad E_{is}^G = E_{is}^F + \underbrace{E_{is}^K}_{\text{Capital}}$$

‣ Back

Equilibrium

- **Equilibrium wages**
  - Clear goods market
    - ⋆ R_{is}: sales across destinations and uses

$$R_{is} = \sum_j \left(\pi^G_{ijs} E^G_{js} + \pi^M_{ijs} E^M_{js} \right)$$

  - Labor markets (Γ^L_{is}: low-skilled task share)

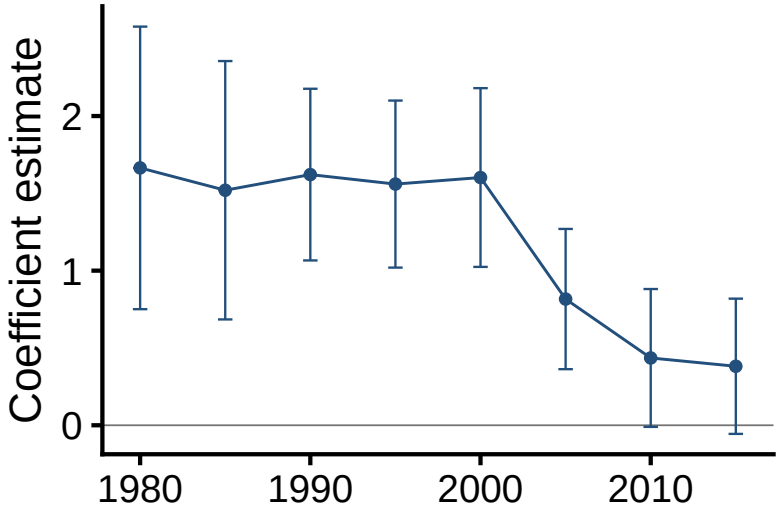
$$w^H_i H_i = \sum_s \alpha^H_{is} R_{is},$$

$$w^L_i L_i = \sum_s \alpha^T_{is} \Gamma^L_{is} R_{is}$$

  - Determine prices, task shares, and trade flows

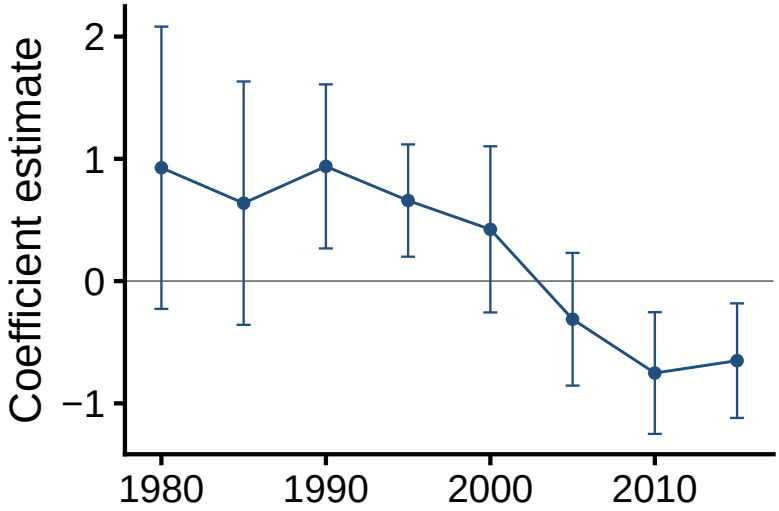
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Exporter-Sector Fixed Effects



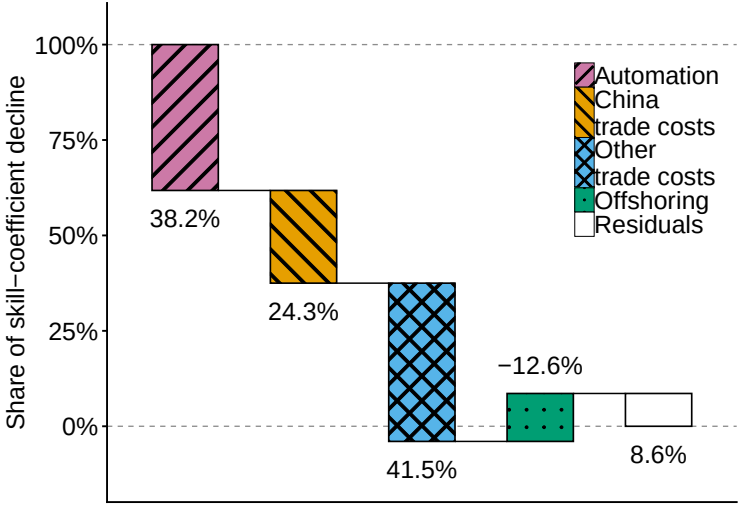
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Romalis (2004): Total Exports



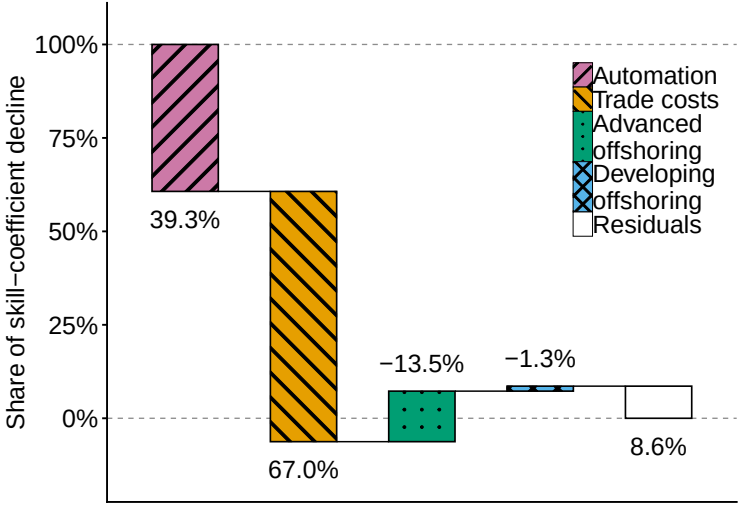
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China and Other Trade Costs



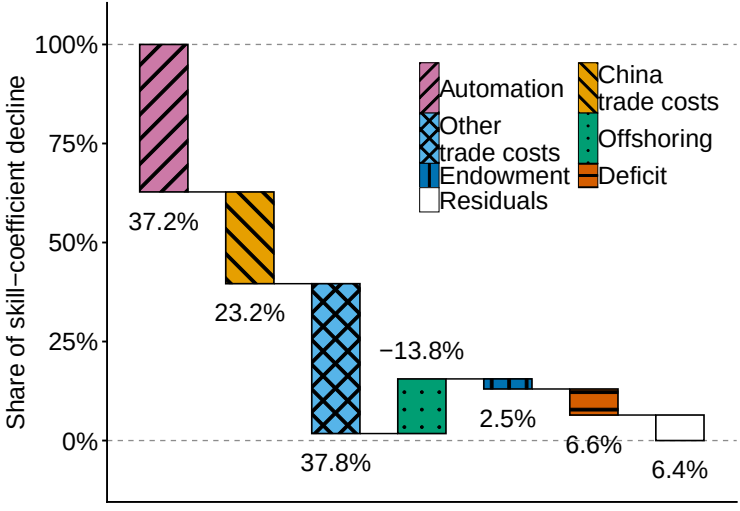
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Offshoring to Advanced and Developing Economies



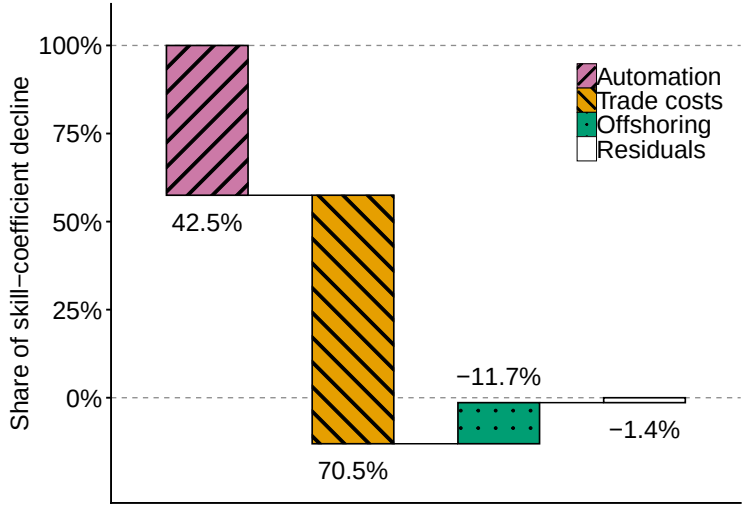
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Adding Endowment and Deficit Changes



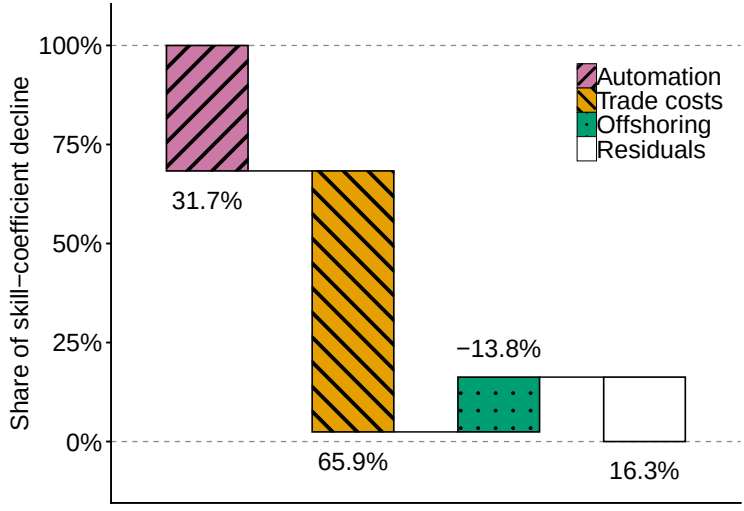
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Weighted Decomposition: 2015 Exporter-Sector Exports



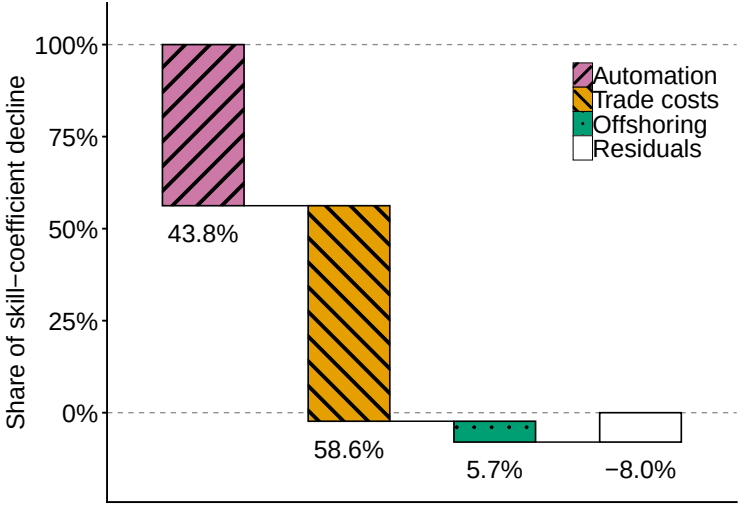
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Alternative Task-Assignment Elasticity: $\varepsilon = 1.0$



‣ Back

Shorter Calibration Window: 1995–2014



‣ Back

Exact Hat Algebra: Tasks

- **Task assignment:** $\varepsilon = 1.4$; substitution elasticity $1 + \varepsilon = 2.4$
  - $\Gamma_{is}^{K,L}$: capital/labor task shares; Ω_{is}': updated capital/labor odds

$$\Omega_{is}' = \frac{\Gamma_{is}^{K}}{\Gamma_{is}^{L}} \left(\frac{\widehat{w}_i^{L}}{\widehat{w}_{is}^{K}} \right)^{\varepsilon}, \qquad \Gamma_{is}^{K\prime} = \frac{\Omega_{is}'}{1 + \Omega_{is}'}$$

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- **Task price** (ε: odds elasticity to w_i^L / w_{is}^K)

$$\widehat{w}_{is}^{T} = \widehat{w}_i^{L} \left(\frac{1 - \Gamma_{is}^{K\prime}}{1 - \Gamma_{is}^{K}} \right)^{1/\varepsilon}$$

Exact Hat Algebra: Sourcing

- **General- and intermediate-use prices**
  - $\widehat{g}^G = \widehat{c}\widehat{\tau}$; $\widehat{g}^M = \widehat{c}\widehat{\tau}\widehat{\chi}$
  - Initial sourcing support fixed

$$\widehat{P}^u_{jr} = \left[\sum_{i \in \mathcal{O}^u_{jr}} \pi^u_{ijr} \widehat{\lambda}_r (\widehat{g}^u_{ijr})^{-\theta} \right]^{-1/\theta}$$

Channel Accounting: From Shocks to the Coefficient

- E^u_{js}: use-u spending; $X^u_{ijs} = \pi^u_{ijs} E^u_{js}$; PPML uses $X^G_{ijs} + X^M_{ijs}$

Channel Accounting: From Shocks to the Coefficient

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$$\mathrm{d}\ln X^u_{ijs} = \underbrace{D^u_{ijs} - \sum_\ell \pi^u_{\ell js} D^u_{\ell js}}_{\text{Direct}} - \underbrace{\theta \left(\mathrm{d}\ln c_{is} - \sum_\ell \pi^u_{\ell js} \mathrm{d}\ln c_{\ell s} \right)}_{\text{Unit cost}} + \underbrace{\mathrm{d}\ln E^u_{js}}_{\text{Expenditure}},$$

where

- Direct: Efficiency term

$$D^u_{ijs} = \mathrm{d}\ln \widehat{\lambda}_s - \theta\, \mathrm{d}\ln \tau_{ijs} - \theta\, \mathbf{1}\{u = M\}\, \mathrm{d}\ln \chi_{ijs}$$

- Unit cost: Productivity, *net* of λ_{is} shifter.
- Expenditure: Expenditure patterns of destinations

Channel Accounting

Channel Accounting by Shocks: Each column reports one shock alone, 1980–2014.

	Trade costs	Automation	Offshoring
Direct	−0.98	0	−0.005
Unit cost	−0.19	−0.60	+0.20
Expenditure	+0.04	−0.19	+0.10
Total (one shock alone)	−1.14	−0.79	+0.29

- Trade costs: Direct (market access) accounts for 86.6%
- Automation: unit costs = task reassignment from low-skilled labor to capital

▸ More on Trade-Cost Channels

Channels: China and Other Trade Costs

Channel	China	Other trade costs	Joint
Market access	−0.99	−0.99	−0.98
Unit cost	+0.32	−0.32	−0.19
Expenditure	+0.02	+0.10	+0.04
Total	−0.66	−1.21	−1.14

- Market access accounts for most of the joint change
- Unit-cost effects have opposite signs

‣ Back

Task Reassignment and Sectoral Reallocation

- **Factor-market clearing**
  - b_{is}^H, b_{is}^L: wage-bill shares; R_{is}: sectoral revenue
  - Γ_{is}^L: low-skilled task share

$$\mathrm{d}\ln\left(\frac{w_i^H}{w_i^L}\right) = \underbrace{\mathrm{d}\ln\left(\frac{L_i}{H_i}\right)}_{\text{Relative supply}} + \underbrace{\sum_s (b_{is}^H - b_{is}^L)\,\mathrm{d}\ln R_{is}}_{\text{Sectoral reallocation}} - \underbrace{\sum_s b_{is}^L\,\mathrm{d}\ln\Gamma_{is}^L}_{\text{Task reassignment}}$$

‣ Back

Welfare: Real Income

- **Representative-household real income**
  - I_i: income; P_i^C: consumption price index

$$W_i = \frac{I_i}{P_i^C}$$

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- **Labor income and trade deficits**

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‣ Back

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- **Consumption price index**
  - $\nu_{ir,1980}$: fixed expenditure shares; P_{ir}^G: general-use prices

$$\widehat{P}_i^C = \exp\left(\sum_r \nu_{ir,1980} \ln \widehat{P}_{ir}^G\right)$$

‣ Back